

Mini SIEM:

Automated Security Event Detection and Analysis

Group 6

Prepared By:

Ahmed Abd Elnasser	21133233
Shahd Ahmed Mohammed Gamel	21125230
Androw Akmal Fadeil	21131006
Esmail Ahmed Esmail Hassan	21130651
Omar Mahmoud Fawzy	21121252

Requirements Gathering

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1. Stakeholders and User Profiles

Stakeholder	Role	Primary Needs
SOC Analyst (Small Org)	Primary User	Fast, automated threat lookups; reduce manual browsing across platforms; export results for reporting
Cybersecurity Student	Primary User	Learn threat intelligence; understand IOC analysis; beginner-friendly guidance
IT Administrator	Secondary User	Quick check of suspicious IPs hitting the firewall; no deep security expertise required
Project Team (6 members)	Developers	Clear technical specifications, defined APIs, realistic milestones
DEPI Reviewers	Evaluators	Academic rigor, proper documentation, working prototype demonstration

2. Functional Requirements

2.1 Input Module

Feature	Description
IP Address Input	User can enter single IPv4 or IPv6 address for analysis
Domain / URL Input	User can enter a domain name or full URL for analysis
File Hash Input	User can enter MD5, SHA-1, or SHA-256 hash for malware lookup
Email Address Input	User can enter an email address for phishing/spam intelligence lookup
Batch Input	User can paste a list of indicators (one per line) for bulk analysis
Input Validation	System validates format before submitting to APIs; shows inline errors for invalid input
Auto-detect Type	System automatically identifies input type (IP vs domain vs hash) without requiring user selection

2.2 Analysis Engine

Feature	Description
VirusTotal Integration	Query VirusTotal API v3 for the submitted indicator; retrieve detection ratio and scan results
AbuseIPDB Integration	Query AbuseIPDB API for IP addresses; retrieve abuse confidence score and category data
AlienVault OTX Integration	Query OTX API for reputation and threat pulse data; retrieve pulse count and IOC categories

2.3 Dashboard and Results Display

Feature	Description
Results Summary Card	Prominent display of Smart Score with color-coded verdict (Safe / Suspicious / Malicious)
Per-API Result Panels	Separate panel for each API showing raw scores, report counts, and key data fields
Threat Category Labels	Display threat categories (e.g., 'Brute Force', 'Phishing', 'Malware') from API reports
Geolocation Display	For IP addresses, display country, ISP, and ASN information
WHOIS Summary	For domains, display registration date, registrar, and expiry date
Export to PDF/CSV	User can export the full analysis report as a PDF or CSV file
Charts / Visualizations	Bar chart or gauge showing detection distribution across antivirus engines (VirusTotal)

2.4 Query History

Feature	Description
History Log	System stores all past queries with timestamps and results summary
Re-run Query	User can click a past query to rerun the analysis with fresh API data
Search History	User can search and filter past queries by indicator, date, or verdict
Clear History	User can delete individual or all history entries

2.5 Help Page

Feature	Description
Glossary	Definitions of key terms: IOC, SIEM, Threat Intelligence, False Positive, etc.
How-To Guide	Step-by-step illustrated guide showing how to analyze an IP, domain, and hash
Real-World Examples	3 worked examples: phishing domain analysis, known malware hash, attacker IP lookup
API Explanation	Simple explanation of what each API is and what its scores mean
FAQ Section	Answers to common beginner questions (e.g., 'What if an IP scores 50/100?')

3. Non-Functional Requirements

3.1 Performance

- Analysis results must be returned to the user within 10 seconds under normal conditions (free-tier API response times considered).
- The dashboard must load in under 3 seconds on a standard broadband connection.
- Parallel API queries must be implemented to avoid sequential delays.
- Batch processing of up to 10 indicators simultaneously must be supported.

3.2 Security

- All user inputs must be sanitized before processing to prevent injection attacks.
- API keys must be stored as environment variables on the backend and never exposed to the frontend or browser.
- HTTPS must be enforced on any deployed instance.
- Rate limiting must be applied to the application's own API endpoints to prevent abuse.
- No raw malware samples will be handled; only hashes and metadata.

3.3 Usability

- The interface must be navigable by a cybersecurity beginner with no prior SIEM experience.
- All threat verdicts must be color-coded: Green (Safe), Yellow (Suspicious), Red (Malicious).
- Error messages must be human-readable and suggest corrective actions.
- The web-application must be responsive and usable on desktop.

3.4 Reliability and Availability

- The web-application must gracefully degrade if one or more external APIs are unavailable.
- The system must display a clear warning if API rate limits are reached.
- Query history must be persisted across browser sessions using a database (not browser localStorage).

3.5 Maintainability

- Code must be modular: each API integration must be in a separate, independently testable module.
- All API keys and configuration values must be externalized in a .env file.
- Unit tests must cover all core API integration functions.
- README documentation must enable a new team member to run the project locally within 30 minutes.

4. User Stories

4.1 SOC Analyst

1. As a SOC analyst, I want to enter a suspicious IP address and immediately see its abuse score and threat categories, so that I can decide whether to block it within 30 seconds.
2. As a SOC analyst, I want to export an analysis report as a PDF, so that I can include it in my incident report.
3. As a SOC analyst, I want to view the history of past queries, so that I can check if a suspicious indicator was previously investigated.

4.2 Cybersecurity Student

4. As a student, I want to understand what each API score means, so that I can learn how to interpret threat intelligence.
5. As a student, I want to see real-world examples of malicious vs. safe indicators, so that I can develop my analytical skills.
6. As a student, I want the interface to explain terms I don't know, so that I can use the tool effectively without prior experience.

4.3 IT Administrator

7. As an IT admin, I want to quickly check if an IP that hit my firewall is known-malicious, so that I can decide whether to create a block rule immediately.
8. As an IT admin, I want to check if a suspicious domain in a phishing email is known-malicious, so that I can confirm and report the phishing attempt.